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Social Implications of the Detection of an Extraterrestrial Civilization

A Report of the Workshops on the Cultural Aspects
of SETI

Held in three sessions:
October 1991, May 1992, and September 1992
at Santa Cruz, California

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Preface

The search for extraterrestrial intelligence (SETI) is an endeavor that, while directly involving only a small number of scientists, appeals to a large audience. Unfortunately for those who wish to learn more about SETI, access to the results of this research is often problematic. SETI papers are frequently published in journals or conference volumes that are either difficult to find, expensive, or both.

Indeed, in some cases, important SETI research is considered to be of such a specialized nature that few publishers will even consider undertaking its dissemination.

In an attempt to ameliorate this problem, the SETI Institute has created the SETI Press. By taking advantage of modern techniques for layout and reproduction, SETI Press can make significant, specialized monographs available to interested parties at low cost.

As its first publication, SETI Press is pleased to offer the present volume dealing with the societal implications of a SETI detection.

The concept for holding these workshops was the initiative of John Billingham, and the detailed planning was carried out in conjunction with Roger Heyns, David Milne, and Vera Buescher. The original preparation of the manuscripts from which this book is derived was made by David Milne. Their ultimate publication by SETI Press is largely due to the initiative and continued interest of John Billingham.

Although the conferences which led to these manuscripts were held in 1991 and 1992 when SETI was still a NASA activity, the ideas and analysis contained herein are still relevant, interesting, and of potentially great import.

*Seth Shostak
Managing Editor, SETI Press
Mountain View, California
June, 1999*

Foreword

This report was written by scholars from many academic disciplines, who assembled to consider a question of unusual significance;

“What would be the cultural, social, and political consequences if NASA’s HRMS project were to succeed at detecting evidence of an extraterrestrial civilization?”

Before October 12, 1992, this question was largely academic. Although a few limited SETI (= Search for Extra-Terrestrial Intelligence) observations had been made prior to that time, these were small in scale, compared with the comprehensive NASA program of observations that started on that date. Long before the startup of the expanded NASA effort, with its improved likelihood of producing a verified detection of an ETI signal, it had been clear to SETI researchers that the social ramifications of such a discovery (and even of the search itself) needed analysis and investigation. That realization prompted NASA to fund and conduct the Workshop on Cultural Aspects of SETI, a far-seeing look ahead at likely consequences of a major scientific discovery, made in advance of that discovery.

The members of the working group were scholars with expertise in many disciplines – history, comparative religion, psychology, sociology, political science, law, education, anthropology, popular media studies, journalism – as well as SETI scientists. They convened on three occasions (October, 1991, and May and September, 1992) for two- and three-day discussions. The group addressed three objectives of importance to NASA:

- 1) to assess likely short- and long-term responses of various societies to a detection of an ETI signal (or non-detection after a long search);
- 2) to identify ways in which NASA (or others) could better inform the public about the research and about various plausible outcomes;
- 3) to recommend areas for further research, intended to clarify the nature of likely consequences of a detection announcement and to identify effective avenues of informing the public.

The working group focussed upon four important areas for assessment and study:

- 1) the history of SETI-related ideas and possible historical analogs to an ETI detection;
- 2) identification of factors that influence human behavior as might apply in a detection situation;
- 3) likely and recommended responses of policy-making institutions to a detection;
- 4) the interactions between informing, educational, and entertainment institutions and the public and NASA Microwave Survey researchers.

The members researched issues in these areas between meetings, communicated their findings with a coordinator in each study area, and wrote drafts of their findings for discussion and refinement at the meetings. This process,

with some finalizing activities, resulted in the present report.

The members of the working group found this to be a more complex and challenging task than seemed apparent at first sight. They completed their work with an expanded list of topics yet to be examined, new questions that invite research, and new possibilities not yet explored. Their report is a beginning, a first look at a scientific investigation with an extraordinary range of implications for humanity's perception of self and cosmos. It is intended as a preliminary description of possible outcomes and (where these cannot be presently foreseen) of research that might clarify our vision of those outcomes – not as a prescription of how the members feel that various publics should respond. The members' recommendations to NASA identify the next logical steps toward ways in which the Agency might productively identify and clarify likely public responses to this, its most intriguing area of research.

SETI Institute, under the able guidance of Tom Pierson (who also participated with the Education/Media team) was key to the smooth functioning of the workshop and production of this Report. This editor particularly wishes to thank Stephen Doyle for sustained good advice, Michelle Murray for help with the manuscripts, and Kathleen Marzano for critically-needed typing assistance.

David H. Milne
August 23, 1992

Workshops of this magnitude could not have been conducted without the hard work and support of many dedicated individuals, to whom we express our thanks. Funding, and much of the advice and foresight that guided the project, were provided by NASA via Headquarters spokesmen Gary Coulter and John Rummel. The formidable logistics of the conferences were handled most skillfully by Vera Buescher, with Amahl Drake providing invaluable support services and assistance during the meetings. Bob Arnold recorded the whole-group working sessions (while participating as a member of the Human Behavior team). Harold Klein presented a stimulating seminar on his interactions with the media during the exciting days when the Viking landers seemed to be transmitting evidence of life on Mars. The organizational support of the

Introduction

From the time when NASA first began to plan a large-scale program of observation of the sky in search of signals from extraterrestrial societies, the Agency has recognized that the social impact of unambiguous evidence of such civilizations would be substantial. Although it is not possible to estimate the likelihood of finding such evidence, the consensus among many investigators is that the probability is not zero. It seemed prudent, therefore, that a serious effort be made by the Agency to anticipate public reactions to such a discovery. With that purpose in mind, a Working Group on Cultural Aspects of SETI was formed, and convened in the summers of 1991 and 1992.

Everyone whose schedule permitted readily accepted the invitation to participate. Although the topic is far afield from the everyday concerns of most of the participants, all immediately regarded the purpose of the effort to be significant and legitimate, and the outcome as important. These positive attitudes were present at the first meeting and became strengthened throughout subsequent meetings and the intervening intervals when the chapters were written.

This Report is the product of the entire membership of that Workshop. Although different groups were the primary contributors to particular chapters, all chapters were subject to advice and editorial suggestions from the membership at large. Although not all would agree with every sentence, a serious and successful effort was made to ensure that the Report is a collective product, substantively reviewed and supported by all.

While the participants tackled their complex

assignment with enthusiasm, it was clear in the discussions that there was also a great deal of caution and humility. The group was being asked to consider questions of enormous complexity, in anticipation of a truly unprecedented event in human history, with very little guidance from history or relevant past experience. The complexity of the world's political organizations, the modest performances of the behavioral sciences in the prediction of human behavior, the enormous freedom of many relevant institutions to choose among many valid (even if sometimes irresponsible) responses, and many other considerations, all make the events following a detection impossible to predict with certainty. Given this formidable challenge, the working group members are prudently tentative in these, their first assessments.

However, appropriately modest claims for the precision of this preliminary analysis do not constitute the whole of our evaluation of its usefulness. We expect that it will prompt historians to suggest additional data and insights from history, and that it will stimulate behavioral scientists to begin analysis of the factors that motivate human responses to great events. We expect that our recommendations will encourage studies that will increase our understanding of the meaning of the existence of ETI's to important segments of the world population – studies that will better inform us about ourselves, even in the absence of an ETI detection.

We believe that we have identified some problems that warrant NASA's careful attention, as the Agency pursues the search. For example, the international implications of the search

(and a detection) are important, and now is the time to prepare scientific and political organizations to maximize the likelihood of constructive popular responses. We expect that an ETI detection will have immediate effects on media and longer-lasting effects on educational institutions. This report suggests ways in which all of their responses to that discovery, and indeed all of humanity's, can be made positive and more effective.

It is our belief that most of our suggestions can have a positive bearing on NASA policy and practice. The single most important realization that emerged from our deliberations is that advance planning for a possible discovery is of critical importance to the Agency. In particular and as part of that preparation, our recommendations call for careful maintenance of trust and credible relations between the Agency and the public, the educational institutions, the national and international scientific communities, and various national and international governmental organizations. We believe that there is much in this report that will help NASA explain the importance and positive significance of this search, for the US and for humanity.

We wrote this Report for NASA Agency officers. However, we have been conscious throughout of a broader audience. As indicated above, we are hopeful that our efforts will be provocative to many disciplines, professions, and organizations: to social scientists, historians, political leaders, scientists, educators, and many others. We are hopeful that our efforts will stimulate insightful new research on their part. And finally, for general readers, we hope that this preliminary exercise will encourage recognition that SETI – an activity with potential for significant impact on all of our lives and perspectives – is not only worth pursuing as a scientific endeavor, but that an actual discovery, should it occur,

would be accepted with equanimity and pleasure.

Roger Heyns
July 8, 1992

Postscript

Early in October, 1993, as this report was first being readied for publication, the US Congress voted to cancel the NASA SETI program, known as the High Resolution Microwave Survey.

All the chapters in this report should now be read with this turn of events in mind.

We have decided to continue with the publication process for a number of reasons. Most important is the general validity of the analysis and findings of the report to any SETI program in the US, and indeed anywhere on this planet. While this is a US publication, the international implications are obvious. They are to be found in every chapter of the report.

Second, the Targeted Search portion of the NASA SETI program is currently being absorbed into the activities of the SETI Institute in Mountain View, California. This non-profit research and education organization has played a major role in SETI over the last ten years, and was responsible for arranging the workshops and implementing the publication of this report. So this major element of the program will now be continued by the SETI Institute with funding drawn from private sources, and not from NASA.

The stimulus to proceed with all the recommendations made in this report, for a thoroughgoing long-term examination of the societal aspects of SETI, remains as important as ever.

Last, but not least, the intensive effort put into the report by all the Workshop participants had already been completed.

We therefore asked the SETI Institute to proceed with their plans for publication, and hope that you will find the report thoughtful, stimulating and enjoyable.

Roger Heyns, John Billingham

IN MEMORIAM

A Tribute to Roger Heyns



Karsh, Ottawa

Roger Heyns, President of the William and Flora Hewlett Foundation and Co-Chairman of the Working Group.

It is with great sadness that I have to relate the untimely death of Roger Heyns on September 11, 1995. He was Co-chair of the Working Group, and guided us through all the complexities we encountered in addressing a subject on which this was basically the first high-level conference.

Roger received his Master's degree in clinical psychology and his Ph.D. in social psychology from the University of Michigan, where he remained as a member of the Psychology Department until 1965. He was Chancellor of the University of California at Berkeley from 1965 to 1971, and then President of the

American Council on Education until 1977. He next became President of the William and Flora Hewlett Foundation.

He was a distinguished scholar in the social psychology and education communities, and received many accolades for his original contributions. He was much in demand as an advisor to Federal and State organizations, the academic community, and to the private sector. Not least, he was on the Board of Directors of the SETI Institute, and we are all in debt to him for his sage advice over the years.

He was one of the few in the academic community who early recognized the immense importance to society of possible future communication with other civilizations in the galaxy, and was prepared to lead us into serious and scholarly Workshop discussions of the subject. The outcome, in the shape of this report, is rich in context and debate, and a model for further conferences which we all recommend take place in the near future.

This document, which is a broad investigation of the cultural aspects of SETI, is a tribute to the leadership, far-reaching vision and deep analytical skills of Roger Heyns. It is important to acknowledge his considerable contribution to this new domain, and express our appreciation for what he did.

John Billingham, July, 1999